LAB 18 RESULTS

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OVERVIEW

Objective

- Understand the principle behind measuring the power rail current
- Use a series resistor, R_{Sense}, in series with the power rail to convert current into a measurable voltage

Measurements

- Measuring the inrush current of three identical circuits but each with a different decoupling capacitor value

FINDINGS

With three identical base circuits but differing decoupling capacitors., it is easy to measure the current flowing into the power rail. Utilizing an oscilloscope and using two single-ended probes to measure the differential voltage, the power rail current can be calculated. It is found that the higher the decoupling capacitance, the larger the current draw on the power rail.

Decoupling Capacitance	Power Rail Current Draw
1 μF	**1x**
10 μF	*2.666x more*
100 μF	*5.111x more*

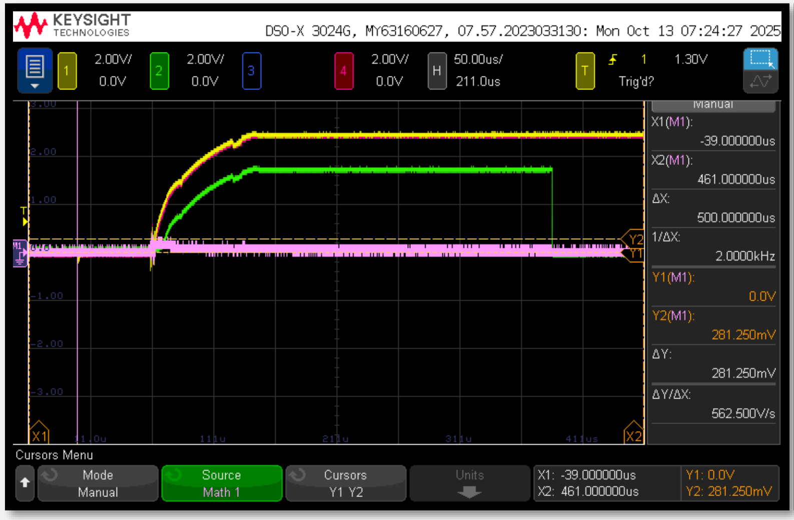
I expected the initial current draw for the 1 μF decoupling capacitor to be higher than it was. But the circuit worked as intended and no discrepancies were found.

FULL MEASUREMENTS

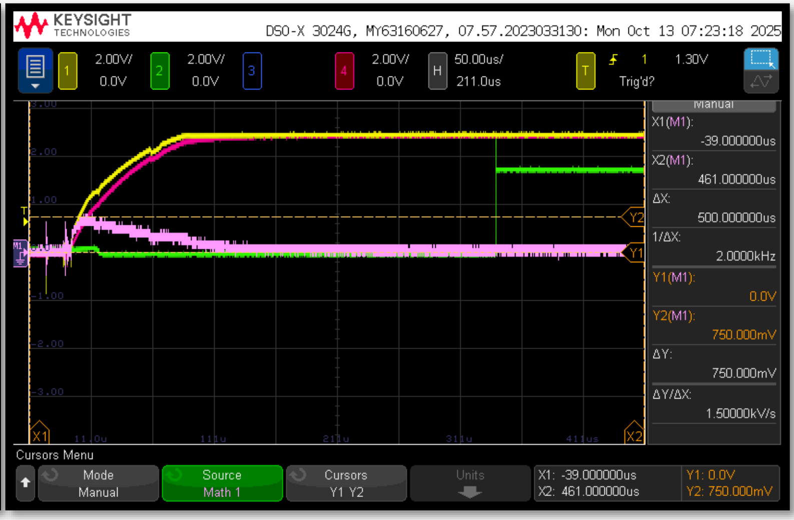
An R_{Sense} value of $1\ \Omega$ is used and is what the differential voltage is measuring across. I chose the smallest available series resistor to try to minimize circuit delay. It is found that the higher the decoupling capacitance, the larger the current draw. A picture of the measured circuit can be observed next to the scope images.

Decoupling Capacitance	Power Rail Current Draw
1 µF (Scope 0)	$\frac{\mathbf{281.250 \times 10^{-3}}\ \boldsymbol{V}}{\mathbf{1\Omega}} = \mathbf{0.28125}\ \boldsymbol{A}$
10 µF (Scope 1)	$\frac{\mathbf{750 \times 10^{-3}}\ \boldsymbol{V}}{\mathbf{1\Omega}} = \mathbf{0.75}\ \boldsymbol{A}$
100 µF (Scope 2)	$\frac{\mathbf{1.43750}\ \boldsymbol{V}}{\mathbf{1\Omega}} = \mathbf{1.4375}\ \boldsymbol{A}$

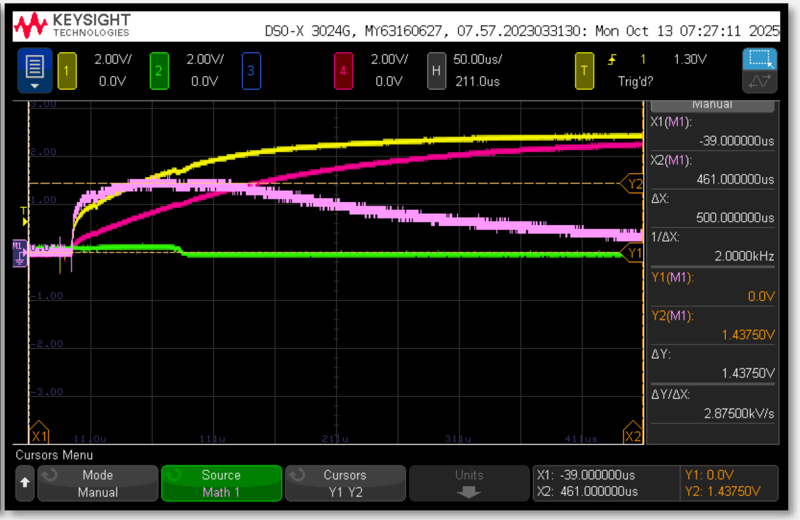
OSCILLOSCOPE/CIRCUIT IMAGES



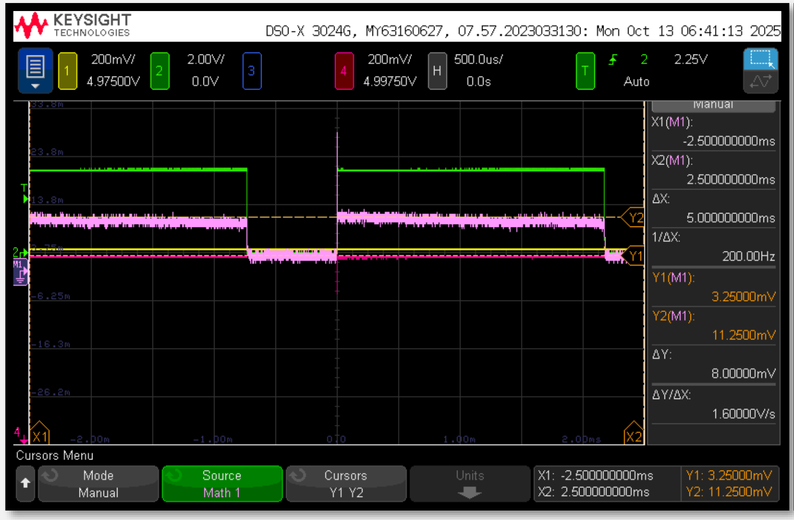
1 μF: Scope 0



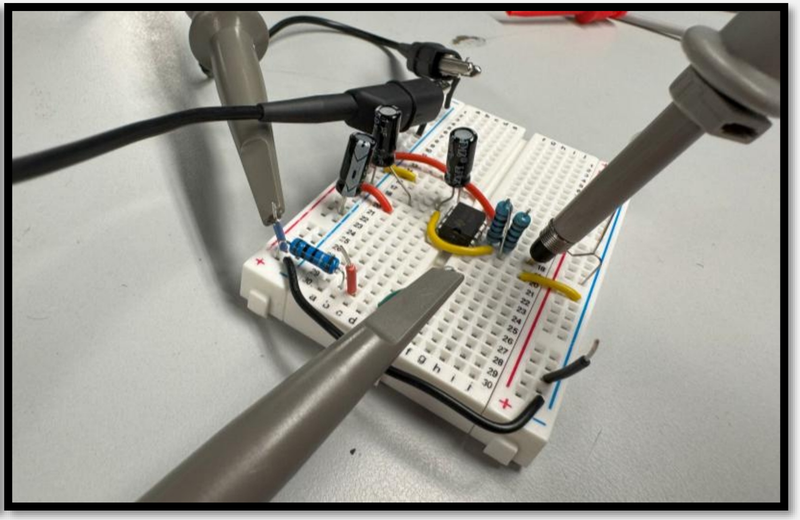
10 μF: Scope 1



100 μF: Scope 2



Differential Voltage Measurement Example: Scope 3



Circuit Image